UNIT 5 · 8 hours

P, PI and PID Controllers



Compensators, Controllers and State-Space Analysis

Control Systems (22EEG53)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

U5 Learning objectives

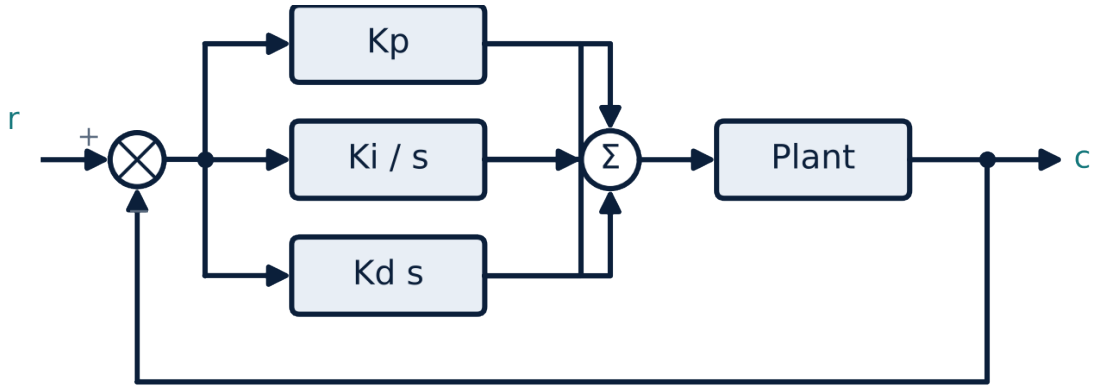
- Write the transfer functions of P, PI, PD and PID controllers
- Explain the effect of each action on the response
- Tune a PID controller by the Ziegler–Nichols method
- Compare PID with lead/lag compensation

Syllabus context

Compensators and controllers: introduction to compensators, lag, lead and lag-lead compensators, P, PI and PID controllers.

U5

PID controller structure



- u(t) = Kp e + Ki ∫e dt + Kd de/dt.
- Gc(s) = Kp + Ki/s + Kd s = Kp[1 + 1/(Ti s) + Td s].
- Ki = Kp/Ti, Kd = Kp Td.
- Parallel branches summed to drive the plant.

U5

Controller actions

Action	Gc(s)	Role
P	Kp	Reduces error and rise time; leaves a steady-state error
I (PI)	Kp(1 + 1/Tis)	Eliminates steady-state error; adds a pole at the origin (slower, less stable)
D (PD)	Kp(1 + Tds)	Anticipates; adds damping; speeds response; amplifies noise
PID	Kp(1 + 1/Tis + Tds)	Combines all three

U5

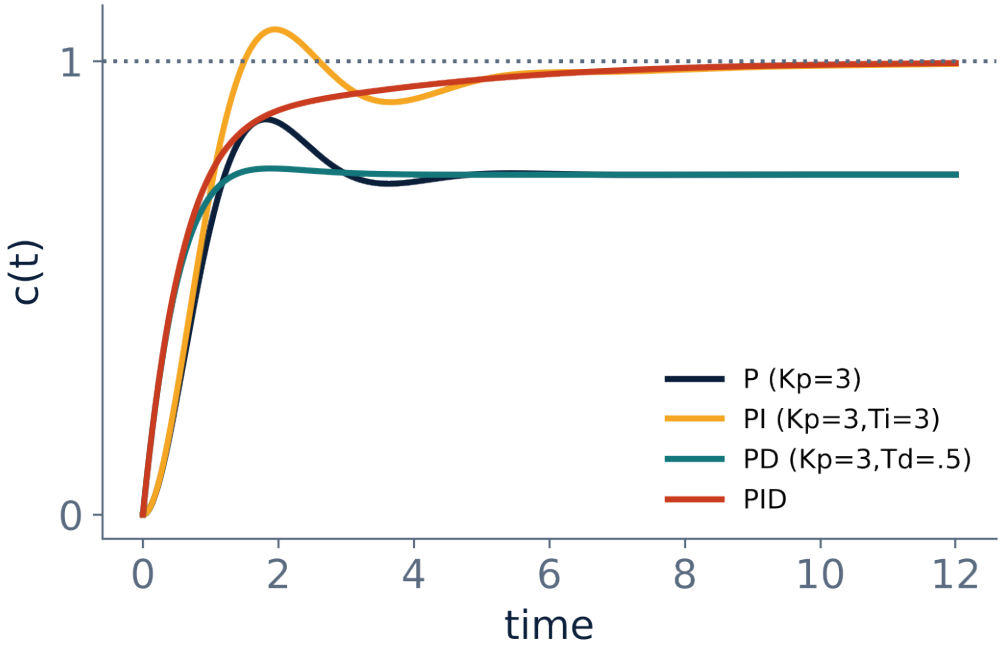
Effect of increasing each gain (typical)

Gain increased	Rise time	Overshoot	Settling time	Steady-state error
Kp	Decrease	Increase	Small change	Decrease
Ki	Decrease	Increase	Increase	Eliminate
Kd	Small change	Decrease	Decrease	No change

Use as a guide for manual tuning; interactions exist.

U5

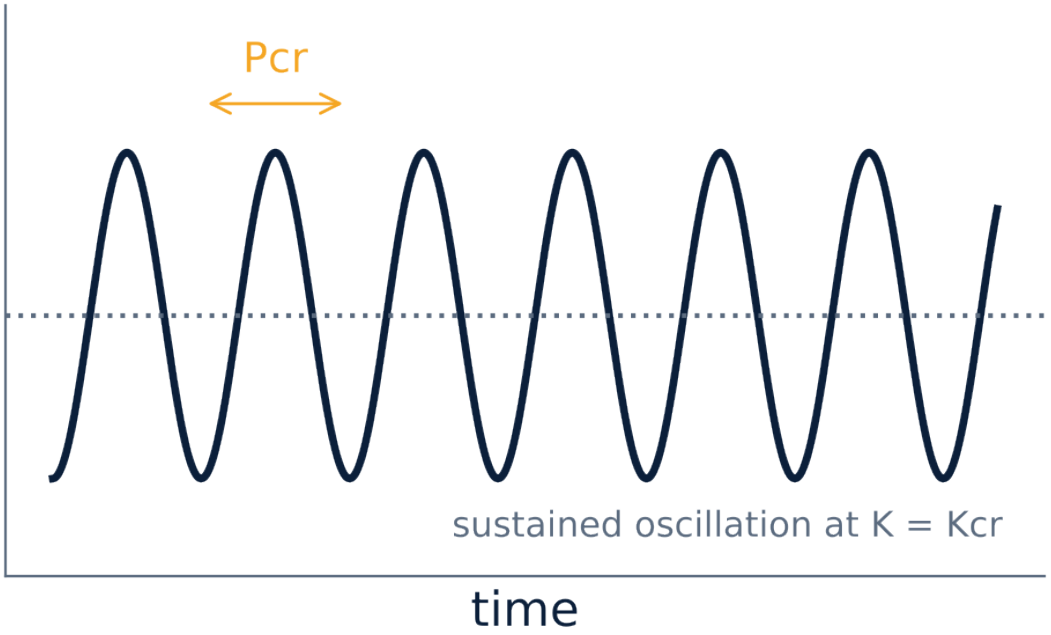
Step responses with P, PI, PD and PID



- P: fast but steady-state error remains.
- PI: error eliminated, some overshoot.
- PD: lower overshoot, faster, error remains.
- PID: best compromise.

U5

Ziegler–Nichols tuning (ultimate gain method)



- Set Ti = ∞, Td = 0; raise Kp until sustained oscillation: Kcr, period Pcr.
- P: Kp = 0.5Kcr.
- PI: Kp = 0.45Kcr, Ti = Pcr/1.2.
- PID: Kp = 0.6Kcr, Ti = 0.5Pcr, Td = 0.125Pcr.

U5

Numerical: ZN tuning

Problem For the plant $K/[s(s+1)(s+2)]$ the loop oscillates at $K = 6$ with $\omega = \sqrt{2}$ rad/s (Routh). Find PID settings.

- $K_{cr} = 6$; $P_{cr} = 2\pi/\omega = 2\pi/1.414 = 4.44$ s
- $K_p = 0.6 \times 6 = 3.6$
- $T_i = 0.5 \times 4.44 = 2.22$ s; $T_d = 0.125 \times 4.44 = 0.555$ s

Answer: $K_p = 3.6$, $T_i = 2.22$ s, $T_d = 0.555$ s ($K_i = 1.62$, $K_d = 2.0$)

U5

Key takeaways

- PID: Kp + Ki/s + Kd s
- P: speed; I: removes steady-state error; D: damping
- ZN: PID Kp = 0.6Kcr, Ti = 0.5Pcr, Td = 0.125Pcr
- PI ≈ lag action; PD ≈ lead action

U5

Practice questions

Q1. Write the transfer function of a PID controller.

Q2. Explain the effect of increasing Ki and Kd.

Q3. Kcr = 10, Pcr = 2 s: find ZN PID values.

Q4. Why does integral action eliminate steady-state error?